RANKS

Ranking, Position in a Row / Queue and Ordering

OAVS TGT | Reasoning (Verbal) | Topic 2 of the syllabus list

Syllabus line (Reasoning, 30 Marks): “The Verbal Reasoning topics like Series, **Ranks**, Direction, Arrangement, Coding, Decoding, Analogy and Classification/Odd Pair, Syllogism and Statement Conclusion.”

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). All 9 rank-type questions found in these papers are solved here.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	Counts, patterns, priority, index of all 9 PYQs
2	Foundation	Picture of the “+1” rule, formula table, language of ranks
3	Six Question Types	Rule, shortcut, step-by-step PYQ solutions
4	Ordering from Comparisons	Heavier / taller / older chains, “who is the heaviest”
5	Shortcuts, Traps and Scope	Mistakes that cost marks; which PYQs belong to Arrangement
6	Practice Set	22 original questions with answer key
7	Quick Revision	One-page cheat sheet

SCOPE: Seating and position puzzles with days, floors, boxes or several people (for example the exam-day and floor puzzles) belong to **Arrangement** and are listed in Chapter 5 so you know where they will be covered. Left-right turns belong to **Direction**.

CHAPTER 1

PYQ Analysis

What the five papers tell us about Ranks

Paper	Q. numbers	Count	What was asked
English 2019	43, 44	2	Total from two ranks; rank from the last
English 2022	39, 54	2	Total from two ranks; position from the other end in a row
Mathematics 2024	none	0	(Only arrangement-type puzzles)
Science 2024	71, 84	2	Total from two ranks (twice)
Social Studies 2023	1, 9, 23	3	Interchange in a line; persons between in a queue; heaviest of six
Total		**9**	**About 2 per paper; found in 4 of the 5 papers**

Pattern	PYQs	Priority
Total = rank from top + rank from bottom − 1	4	★★★ Must
Rank / position from the other end (total known)	2	★★★ Must
Interchange of two positions in a line	1	★★ High
Persons between two people in a queue	1	★★ High
Order by comparison (heaviest / lightest)	1	★★ High
Expected, not yet asked: rank changes, join / leave, middle student, equal rank from both ends	0	★ Expected

PYQ INSIGHT: The formula **total = top + bottom − 1** was asked **four times**: English 2019 Q43, English 2022 Q39 and **both** Science 2024 questions (Q71 and Q84). Every PYQ answer in this topic comes from a one-line formula once you set it up correctly.

PYQ INSIGHT: Rank questions carry **extra information on purpose** (SST 2023 Q1 gives Ali's position, which is never used). Decide which facts you need before calculating.

Index of the 9 PYQs

Paper / Q	Question in short	Ans	See
Eng 2019 Q43	22nd from top, 25th from last: class strength	46	3.1
Eng 2019 Q44	42nd in a class of 70: rank from last	29	3.2
Eng 2022 Q39	112th from top, 344th from bottom: total runners	455	3.1
Eng 2022 Q54	56th from the right in a row of 100: position from left	45th	3.2
Science 2024 Q71	7th from top, 34th from bottom: students	40	3.1
Science 2024 Q84	9th from top, 25th from bottom: students	33	3.1
SST 2023 Q1	115 in a line; Parth and Yug interchange: persons right of Parth	64	3.4
SST 2023 Q9	Queue of 105; persons between Charmie and Manya	42	3.3
SST 2023 Q23	Six rods with weight relations: heaviest	R	Ch. 4

CHAPTER 2

Foundation

The “+1” rule and the formulas behind every question

2.1 Why “top + bottom − 1”

A person who is 5th from the top and 4th from the bottom in a line of 8 is counted **twice** if you add the two ranks. Blue = counted from the top, orange = counted from the bottom, green = the person (counted in both).

From top	1	2	3	4	5	6	7	8
From bottom	8	7	6	5	4	3	2	1

5 + 4 = 9 counts the person twice, so the real total = 9 − 1 = **8**. This one picture explains every formula below.

2.2 Language of ranks

Phrase	Meaning
From top = from front = from the first = from the left (when the line is read left to right)	Counted from the start; same as ‘rank’ in a class
From bottom = from last = from the end = from the right (in the same reading)	Counted from the other end
X is 10th from the top	**9 people are above X** (rank − 1), X is the 10th person
Rank 1 = topmost; rank improves as the number decreases	A rise of 3 places: 15th → 12th

2.3 Formula table (memorise all)

Situation	Formula	Example
Same person, ranks from both ends	Total = a + b − 1	22 + 25 − 1 = 46
Rank from the other end	Other = Total − rank + 1	70 − 42 + 1 = 29
Persons above / below a person	Above = rank − 1; below = Total − rank	12th of 40: 11 above, 28 below
Persons between two people (same end, ranks m < n)	n − m − 1	15th and 20th: 4 between
Persons between, X a-th from top and Y b-th from bottom (X above Y)	Total − a − b	40 − 9 − 13 = 18
Middle person, Total odd	(Total + 1) ÷ 2	49: 25th
Same rank r from both ends	Total = 2r − 1	20th from both: 39

CHAPTER 3

Six Question Types

Rule, shortcut and step-by-step PYQ solutions

3.1 Type 1: Find the total [★★★ four PYQs]

Spot: one person, two ranks from opposite ends. **Rule:** Total = a + b − 1.

PYQ Eng 2019 Q43 Class strength

Q. Samanvi ranks 22nd from top and 25th from last. What is the strength of the class?

Step 1. Rank from top = 22, rank from last = 25.
Step 2. Total = 22 + 25 − 1 = 46.
Answer: 46 (Option D)

PYQ Eng 2022 Q39 Same pattern, large numbers

Q. X is 112th from the top and 344th from the bottom. Total marathon runners?

Step 1. Total = 112 + 344 − 1.
Step 2. 112 + 344 = 456, minus 1 = 455.
Answer: 455 (Option B)

PYQ	Given	Working	Answer
Science 2024 Q71	Hemant 7th top, 34th bottom	7 + 34 − 1	**40**
Science 2024 Q84	Karan 9th top, 25th bottom	9 + 25 − 1	**33**

SHORTCUT: Add the two ranks, then subtract 1. The answer is always **one less than the sum**, so options equal to the plain sum (47, 457, 41, 34) are the trap options.

3.2 Type 2: Rank / position from the other end [★★★]

Rule: Other = Total − given + 1. It works for classes (top and bottom) and rows (left and right).

PYQ Eng 2019 Q44 Rank from last

Q. Ravali ranks forty second in a class of seventy students. What is her rank from last?

Step 1. Total = 70, rank from top = 42.
Step 2. Rank from last = 70 − 42 + 1 = 29.
Answer: 29 (Option C)

PYQ Eng 2022 Q54 Row of cards

Q. Mohit selected the 56th card from the right in a row of 100 cards. What is its position from the left?

Step 1. Total = 100, from right = 56.
Step 2. From left = 100 − 56 + 1 = 45.
Answer: 45th (Option B)

TRAP: 70 − 42 = 28 is the number of people **below** her; her own rank is one more, 29. Forgetting the +1 gives the wrong option 28 (listed in the paper).

3.3 Type 3: Persons between two people [★★]

Rule (same end): ranks m and n (m < n) → between = n − m − 1. **If the ranks are from opposite ends** (a from front, b from back, front person first): between = Total − a − b.

PYQ SST 2023 Q9 Queue of 105 (one number lost in the source paper)

Q. In a queue of 105, only 28 are before Charmie. Some number of persons are before Manya. How many are between Charmie and Manya?

Step 1. Charmie is 29th from the front (28 before her).
Step 2. Between = (persons before Manya) − (persons before Charmie) − 1.
Step 3. The official answer 42 corresponds to 71 persons before Manya: 71 − 28 − 1 = 42.
Answer: 42 (Option A)

NOTE: The printed question in the source file has a blank where Manya's number should be, so only the method is reliable here. Convert every 'before' count into a position first (before + 1), then use **difference − 1**.

SHORTCUT: People between two ranks are always the **difference minus 1**, never the plain difference. Ranks 15th and 20th: 20 − 15 = 5, minus 1 = 4 between.

3.4 Type 4: Interchange of positions [★★]

Rule: after A and B swap, A stands exactly where B stood, so A's new position equals B's old position (and vice versa). Find the answer from the **person who moves into the spot**.

PYQ SST 2023 Q1 115 students in a line, same direction

Q. Only 33 students are to the right of Ali. 12 are to the left of Parth. Yug is 65th from the right end. If Parth and Yug interchange, how many students are to the right of Parth?

Step 1. After the swap Parth stands where Yug stood: 65th from the right.
Step 2. Students to the right of Parth = 65 − 1 = 64.
Step 3. Ali's data (33 to the right) and Parth's old position (12 to the left) are not needed; they are distractors.
Answer: 64 (Option 3)

SHORTCUT: Do not recalculate the whole line. After a swap, ask only: **which spot does this person now occupy?**

3.5 Type 5: Rank changes, joining and leaving [★ expected]

Change	Rank from top	Rank from bottom	Total
k students join above the person	+ k	same	+ k
k students join below the person	same	+ k	+ k
k students above leave	− k	same	− k
k students below leave	same	− k	− k
Person's rank improves by k places	− k	+ k	same
Person's rank drops by k places	+ k	− k	same

SHORTCUT: Whoever changes **on one side** of the person leaves the rank from the **other** end unchanged. Example: class of 40, Amit 12th from top (29th from bottom). If 5 students above him leave, he is 7th from top and still 29th from bottom.

3.6 Type 6: Two different persons, unknown total [★ expected]

Draw a line: X at position a from the left, then n people, then Y, who is b from the right. Y's position from the left = a + n + 1, so **Total = a + n + b**.

Example: P is 14th from left, Q is 19th from right, 8 people between them. Total = 14 + 8 + 19 = **41**.

CHAPTER 4

Ordering from Comparisons

Heaviest, tallest, oldest: build a chain

Step	Action
1	Turn each statement into a symbol: ‘A is heavier than B’ → A > B. Convert ‘half of’, ‘twice’ and ‘5 kg less’ into formulas using a letter.
2	Link statements through a **common person** to build one chain (transitivity: A > B and B > C ⇒ A > C).
3	If two chains meet only at the top or the bottom, the people on different branches **cannot be compared**.
4	Answer: heaviest = start of the chain, lightest = end. If a person is not provably above everyone else, the answer is ‘cannot be determined’.

PYQ SST 2023 Q23 Six iron rods P, Q, R, S, T, U (all different weights)

Q. P is half of Q. R is twice S. T is 5 kg less than R. U is some kg less than S. Q is half of S. Which is the heaviest?

Step 1. Let S = 2x. Then Q = x (half of S) and P = x/2 (half of Q).
Step 2. R = 2S = 4x. T = R − 5 = 4x − 5, which is smaller than R. U = S − some, which is smaller than S.
Step 3. Now S = 2x < R = 4x, and Q, P, U, T are all below R. R is above every other rod.
Answer: R (Option 2)

SHORTCUT: To find the heaviest, you do not need numbers: show that one rod is above **every** other. Here R is above S (twice), above T (5 kg less) and S is above Q, P and U.

TRAP: Equal-looking relations (A = twice B, C = twice B) make A and C **equal**. If the question says ‘all different’, that cannot occur. Check the statement before you choose between two close candidates.

Combining two chains

Statements	Chain	Can be answered	Cannot be answered
P > Q, R > Q, S > P, S > R	S > (P, R) > Q	Tallest S, shortest Q	P vs R
A > B > C, D < C, D > E	A > B > C > D > E	Heaviest A, lightest E, middle C	—

CHAPTER 5

Shortcuts, Traps and Scope

Marks saved and marks lost

5.1 Shortcuts

- **Two ranks of the same person → add and subtract 1.** Do it mentally in 5 seconds.
- **Option elimination:** the answer is never the plain sum of the two ranks, so remove that option first.
- **Other end:** Total − rank + 1. Check: the rank from the two ends must add up to Total + 1 (42 + 29 = 71 = 70 + 1).
- **Interchange:** only the person who moves matters; read off the new spot.
- **Between:** difference minus 1; or Total − a − b when the ranks are from opposite ends.
- **Draw a line** of boxes whenever two different persons are involved. A 10-second sketch avoids off-by-one errors.

5.2 Traps

Trap	How to avoid it
Forgetting the −1 / +1	Test with a small case: 1st from top and 1st from bottom in a line of 1 → 1 + 1 − 1 = 1.
Using ‘between’ without subtracting 1	15th and 20th have 4 people between, not 5.
Extra data	Ali’s position in SST 2023 Q1 is not needed. Mark the data you actually use.
Mixing ‘before’ and ‘position’	28 before Charmie means she is 29th.
Left and right reading	‘From the right’ is the other end of ‘from the left’. In a line facing north, the person’s left is west; use the diagram.
Assuming ties	Ranks are all different unless the question says otherwise.

5.3 Scope: PYQs that look like rank questions but belong to other topics

PYQ	Type	Covered in
Maths 2024 Q82, Q84; Science 2024 Q95	Exam days (Monday to Sunday) ordering	Arrangement
Science 2024 Q86; SST 2023 Q8	Floors of a building	Arrangement
SST 2023 Q28	Boxes one above the other	Arrangement
Eng 2022 Q34 and Q43; Science 2024 Q88; SST 2023 Q6	Linear seating with left / right	Arrangement
Eng 2022 Q37, 42, 48, 49, 55; Science 2024 Q93; SST 2023 Q27	Circular seating	Arrangement
Eng 2019 Q50, Q57; Eng 2022 Q46, Q60; Maths 2024 Q86; Science 2024 Q91; SST 2023 Q16, Q19, Q21	Directions, clocks and distance	Direction

CHAPTER 6

Practice Set

22 original questions; answers on the next page

Q	Question
1	Rohan ranks 15th from the top and 22nd from the bottom in a class. How many students are there?
2	Meena is 18th from the top in a class of 50. What is her rank from the bottom?
3	In a row of 60 people Arun is 25th from the left. What is his position from the right?
4	A boy is 12th from the top and also 12th from the bottom. How many students are there?
5	In a class of 40, Kiran is 9th from the top and Sunil is 13th from the bottom (Kiran is above Sunil). How many students are between them?
6	P is 14th from the left and Q is 19th from the right in a line. 8 persons are between them (P is to the left). How many persons are in the line?
7	In a class of 45, Ravi is 17th from the top. Priya is 8 ranks below Ravi. What is Priya’s rank from the bottom?
8	A is 11th from the left and B is 14th from the right in a row. If they interchange, A becomes 20th from the left. How many persons are in the row?
9	In a line of 120 students, Ali is 30th from the left and Beena is 40th from the right. After they interchange, how many students are to the right of Ali?
10	Neha was 12th from the top in a class of 40. After a test her rank drops by 5 places. What is her rank from the bottom now?
11	Tom is 10th from the top in a class of 35. 6 new students join, all with marks higher than Tom. What is Tom’s rank from the bottom now?
12	In a class of 49, what is the rank from the top of the middle student?
13	A student is 20th from the top and 20th from the bottom. How many students are there?
14	In a test, 6 students scored more than Aman and 17 scored less. How many students took the test?
15	In a queue of 70, Dev is 15th from the front and Ela is 20th from the back (Dev is in front of Ela). How many are between them?
16	Raj is 7th from the left and Sam is 10th from the right. After they interchange, Raj is 12th from the left. How many persons are between them now?
17	Seema is 10th from the top. Rita is 5 ranks above Seema and is 21st from the bottom. How many students are there?
18	In a row of 31, Mohan is 9th from the left and Sohan is 7th from the right. How many persons are between them?
19	In a class of 40, Amit is 12th from the top. If 5 students above him leave, what is his rank from the bottom?
20	A > B, B > C, D < C, D > E (all are weights). Who is the heaviest, and who is the third lightest?
21	P is taller than Q. R is taller than Q. S is taller than P. S is taller than R. Who is the tallest? Can we say who is taller between P and R?
22	A is twice as heavy as B. B is twice as heavy as C. D is lighter than A but heavier than B. Who is the heaviest and who is the lightest?

Answer key with logic

Q	Answer	Logic
1	36	15 + 22 − 1
2	33	50 − 18 + 1
3	36	60 − 25 + 1
4	23	12 + 12 − 1
5	18	Total − a − b = 40 − 9 − 13
6	41	Q from left = 14 + 8 + 1 = 23; total = 23 + 19 − 1
7	21	Priya from top = 17 + 8 = 25; from bottom = 45 − 25 + 1
8	33	A takes B’s old spot: 20th from left; B was 14th from right; total = 20 + 14 − 1
9	39	Beena is 120 − 40 + 1 = 81st from left; Ali takes that spot; right of it = 120 − 81
10	24	New top rank 17; 40 − 17 + 1
11	26	Students joined above him; bottom rank unchanged: 35 − 10 + 1
12	25th	(49 + 1) ÷ 2
13	39	20 + 20 − 1
14	24	6 + 17 + 1 (Aman himself)
15	35	70 − 15 − 20
16	4	Raj is 12th, Sam takes Raj’s old spot (7th) from left; between = 12 − 7 − 1
17	25	Rita 10 − 5 = 5th from top; total = 5 + 21 − 1
18	15	Sohan from left = 31 − 7 + 1 = 25th; 25 − 9 − 1
19	29	Rank from bottom unchanged: 40 − 12 + 1
20	A; C	Order A > B > C > D > E. Third lightest: E, D, C
21	S; no	S > (P, R) > Q; P and R are on different branches
22	A; C	A = 4c, B = 2c, C = c; D between B and A

CHAPTER 7

Quick Revision

One page for the last day

Question asks	Use	PYQ check
Total from two ranks (same person)	a + b − 1	22 + 25 − 1 = 46; 112 + 344 − 1 = 455
Rank from the other end	Total − rank + 1	70 − 42 + 1 = 29; 100 − 56 + 1 = 45
Persons above / below	rank − 1 / Total − rank	12th of 40: 11 above, 28 below
Persons between (same end)	difference − 1	15th and 20th: 4
Persons between (opposite ends)	Total − a − b	40 − 9 − 13 = 18
After interchange	Read the spot the person moves into	SST 2023 Q1: 65th from right → 64 on the right
Joining / leaving	Other-end rank is unchanged	Amit: 12th → 7th top, still 29th bottom
Middle student	(Total + 1) ÷ 2	49 → 25th
Heaviest / tallest	Build the chain; show one above all	SST 2023 Q23: R

The four PYQ formulas at a glance

Total (a + b − 1)	Other end	Interchange	Between
Eng 2019 Q43 → 46 Eng 2022 Q39 → 455 Science 2024 Q71 → 40 Science 2024 Q84 → 33	Eng 2019 Q44 → 29 Eng 2022 Q54 → 45th	SST 2023 Q1 → 64	SST 2023 Q9 → 42 (difference − 1)

ROUTINE: Exam-day routine: sketch a line of boxes, mark every person with a position from one end, convert all other counts to the same end, then apply +1 / −1 carefully. 30 seconds per question.